

Robin CR

Chennai, Tamil Nadu

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Personal Profile

4+ years experienced Embedded Software Developer with proven success in building algorithms. Highly adept at C++ application development, firmware development, ADAS and multi-threading. Passionate engineer with the ability to apply techniques and algorithm development to solve real world problem.

Work Experience

Automotive Robotics India (ARi)

Senior Software Engineer

Chennai, Tamil Nadu

January 2025 – Present

Client: CATERPILLAR (Onsite)

LIDAR Perception for Surface Hauling truck

- Engineered Moving Obstacle Detection solutions leveraging lidar sensor.
- Built and optimized Range image-to-point cloud converters, enabling support for multiple lidar types.
- Developed boundary and berm detection feature for mine site.
- Supported the development of the rut detection feature for haul trucks.
- Integrated open-source Point Cloud libraries into C++ projects to accelerate development.
- Worked in Path planning, Sensor fusion and SLAM algorithms.
- Drove KPI improvements by investigating and resolving root causes of KPI mismatches.
- Developed robust backend systems for a playback tool.

Bosch Global Software Technologies

Embedded Software Engineer

Coimbatore, Tamil Nadu

August 2022 – December 2024

Interior Monitoring System (IMS) – Camera Perception

- Developed algorithms for driver drowsiness module in C++ following design patterns and clean code.
- Developed internal tools for simulating driver behaviors for debugging the issues without hardware.
- Contributed to the development of hardware and software reset for camera.
- Modules implementations such as interfaces, precondition handlers, error handlers, schedulers etc.
- Successfully addressed and resolved Level 8 and 9 QAC++ warnings in the codebase.
- Worked in the project migration on virtual machine to docker.
- Able to handle real world bugs which require proper strategies, Problem Solving and analytical skills

Digital Instrument Cluster

- Worked in all phases of Agile Scrum methodology (Requirement - Design - Coding - Testing).
- Contributed to the development and integration of the Trip Computer module.
- Identified and resolved a critical tell-tale toggling issue.
- Investigated, raised, and fixed a bootloader timeout issue in the ECU.
- Writing unit tests and performs functionality testing to ensure software functionality.
- Implemented an algorithm that will modify the fixed parameters in RTE layer into a configurable in application layer.
- Base level knowledge on Diagnostics, Bootloader and RTE layer is gained.

Gadgeon Systems Pvt Ltd

Embedded Software Developer

Kochi, Kerala

July 2021 - August 2022

- Developed codecs for different sensors using C programming language in a Linux OS environment. (Gateway Programming)
- Developed custom protocols to communicate with different sensors in embedded C.
- Platform dependent and independent code development on different ARM based microcontrollers.
- Interfaced accelerometer, gyroscope, and GPS sensors (Bare metal programming).

Projects

- **Calendar and Clock** : A project which addresses all the problems related to calendar and clock.
- **CPU-GPU Selector** : Implemented a system that decides the model training to be done on CPU, GPU, or GPU on cloud.

Skills

Microcontrollers	ARM Cortex M3 LPC1768, ESP8266
Architecture	Classical Autosar, Adaptive Autosar
Lidar	Velodyne, Luminar, Ouster
Protocols	UART, I2C, SPI, CAN
Design	Enterprise Architecture, PlantUML
Debugger	Valgrind, GDB, Lauterbach Trace32
Tools	CANoe
Unit Testing	Gtest, BOOST
Version control system	Git, RTC, BitBucket
Build	Cmake, Sconsript
AI tools	Github Copilot

Competencies

- **Data Structures**
Array, Linked list, STL library.
- **Embedded C programming**
State machine, Bit wise programming, handshaking mechanisms.
- **C++ Application development**
Move Semantics, Smart Pointers, Templates, Functional Programming, Atomics.
- **AI Integration**
Synthetic dataset generation, Local LLM integration

Certifications

Embedded System Professional

Translot, Kochi

- Solid foundation on basics of embedded C programming.
- Completed the projects based on EEPROM and display systems.
- Working knowledge on POSIX programming is achieved.
- Proper understanding of Linux Fundamentals (System calls, Booting, file system) is gained.

Professional C++ and Clean code

Bosch, Coimbatore

- Leveraged the OOPs and SOLID principles with examples.
- Discussed common pitfalls in C++ for embedded software development.
- Covered Design Patterns explaining their roles in code reusability and scalability.
- Clean coding guidelines and commons practices were addressed.

Education

Rajagiri School of Engineering and Technology

Bachelor of Technology in ECE

Completed with a GPA of 8.2

Kochi, Kerala

Aug 2017 - Jul 2021